

# Inseok Kong

Undergraduate Researcher · Department of GeoInformatics & AI · University of Seoul

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## RESEARCH INTERESTS

I am broadly interested in **Trustworthy Artificial Intelligence**, with a primary focus on **Computer Vision**. My research has primarily centered on **Test-Time Adaptation (TTA)** to enable vision models to reliably adapt to real-world distribution shifts and environmental corruptions. Recently, I am expanding my research focus toward **Uncertainty Calibration** and object hallucination mitigation in vision-language models, striving to build reliable, well-calibrated, and robust visual intelligence systems.

## EDUCATION

### University of Seoul

B.S. in GeoInformatics and Artificial Intelligence (Double Major)

Mar 2022 – Present

Seoul, South Korea

- **Cumulative GPA:** 3.8 / 4.5
- **Relevant Coursework:** Machine Learning, Deep Learning, Pattern Recognition, Data Structures, Algorithms.

## PUBLICATIONS & PREPRINTS

[C: Conference, S: Submitted]

- [C.1] **APC: Transferable and Efficient Adversarial Point Counterattack for Robust 3D Point Cloud Recognition** Mar 2026  
Geunyoung Jung, Soohong Kim, **Inseok Kong**, and Jiyoung Jung Denver, CO, USA  
*IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings (CVPR Findings)*
- [C.2] **MAMVI: 3D Test-Time Adaptation via Masked Multi-View Point Clouds** Jul 2026  
**Inseok Kong**, Geunyoung Jung, and Jiyoung Jung Lyon, France  
*IEEE/IAPR International Conference on Pattern Recognition (ICPR) (Oral Presentation)*
- [S.1] **SATOR: Backpropagation-Free 3D Test-Time Adaptation via Sample-Adaptive Token Routing** Submitted  
**Inseok Kong**, Geunyoung Jung, and Jiyoung Jung  
*Under Review*
- [S.2] **Mitigating Feature Collapse for Test-Time Adaptation of 3D VLMs via Distribution-Guided Regularization** Submitted  
Geunyoung Jung, **Inseok Kong**, and Jiyoung Jung  
*Under Review*

## RESEARCH EXPERIENCE

### Speech & Interactive AI Laboratory (SIA)

Undergraduate Research Student (Advisor: Prof. Heeseung Kim)

Aug 2026 – Present

University of Seoul, South Korea

- Conducting research on **Audio-Vision-Language Models (AVLLMs)**, focusing on hallucination mitigation to enhance multimodal reliability and trustworthiness.

### Statistical Inference and Information Theory Lab (SIIT)

Undergraduate Research Student (Advisor: Prof. Junmo Kim)

Jul 2026 – Aug 2026

KAIST, Daejeon, South Korea

- Conducting individual research on **Large Vision-Language Models (LVLMs)**, focusing on mitigating object hallucination and improving uncertainty calibration to address visual-textual misalignment and enhance multimodal trustworthiness.

### Robotics and Computer Vision Lab (RCV)

Undergraduate Research Student (Advisor: Prof. Jiyoung Jung)

Jul 2025 – Jun 2026

University of Seoul, South Korea

- Researched **3D Test-Time Adaptation (TTA)** under severe domain shifts, investigating lightweight online adaptation strategies for 3D point cloud networks to maintain high inference accuracy while eliminating backpropagation overheads.
- Investigated **Point Cloud Robustness & 3D Adversarial Defense**, studying structural vulnerabilities of 3D point cloud classification architectures and exploring input-level defense mechanisms to neutralize point perturbations dynamically.
- Explored robust representation alignment for **3D Vision-Language Models (VLMs)**, addressing feature collapse and cross-modal misalignment issues under environmental corruptions and out-of-distribution (OOD) scenarios.

PROJECTS

|                                                                                                                                                                                                                                |                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>LG Aimers 8th Cohort</b><br><i>Track: LLM Compression</i>                                                                                                                                                                   | <i>Jan 2026 – Feb 2026</i><br>Online             |
| <ul style="list-style-type: none"><li>Researched and applied model compression &amp; optimization techniques (pruning, quantization) on EXAONE Large Language Models.</li></ul>                                                |                                                  |
| <b>Optimal Autonomous Shuttle Route Planning</b><br><i>Capstone Design Project @ Seoul Big Data Campus</i>                                                                                                                     | <i>Oct 2025 – Jan 2026</i><br>Seoul, South Korea |
| <ul style="list-style-type: none"><li>Developed optimal route planning algorithms using mobility big data to support customized autonomous shuttle services for transportation-vulnerable elderly citizens in Seoul.</li></ul> |                                                  |
| <b>Sportier (UOS Entrepreneurship Club)</b><br><i>Lead Developer @ President (Tools: React.js, JavaScript)</i>                                                                                                                 | <i>Mar 2025 – Jul 2025</i><br>Seoul, South Korea |
| <ul style="list-style-type: none"><li>Led the entrepreneurship club as president and developed a sports matchmaking web application featuring location-based matching and competitive tier rankings.</li></ul>                 |                                                  |
| <b>Football Big Data Analytics</b><br><i>Football Big Data Camp</i>                                                                                                                                                            | <i>Feb 2025</i><br>Seoul, South Korea            |
| <ul style="list-style-type: none"><li>Investigated practical applications of LLMs and Retrieval-Augmented Generation (RAG) in football performance analytics.</li></ul>                                                        |                                                  |

HONORS & AWARDS

|                                                                                                               |                                       |
|---------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Excellence Award</b><br><i>Seoul RISE AI Competition (Project: SATOR)</i>                                  | <i>Jun 2026</i><br>Seoul, South Korea |
| <b>Grand Prize (Seoul Mayor Award)</b><br><i>Seoul Metropolitan Big Data Campus Competition</i>               | <i>Oct 2025</i><br>Seoul, South Korea |
| <b>Excellence Award</b><br><i>UOS Capstone Design Project</i>                                                 | <i>Jan 2026</i><br>Seoul, South Korea |
| <b>Encouragement Award</b><br><i>AICOSS Industry-Academia Hackathon (Diffusion @ LLM Storybook)</i>           | <i>Aug 2025</i><br>Seoul, South Korea |
| <b>Dean’s Award of College of Engineering</b><br><i>AICOSS Deep Learning Bootcamp (2D ViT Classification)</i> | <i>Jan 2025</i><br>Seoul, South Korea |

VOLUNTEER & MENTORING

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|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <b>KB Raschool 3rd Generation Mentor</b><br><i>Samsung Sori-Saem Welfare Center</i>                                                                      | <i>Mar 2025 – Feb 2026</i><br>Seoul, South Korea |
| <ul style="list-style-type: none"><li>Provided tailored mathematics mentoring and academic counseling for hearing-impaired 9th-grade students.</li></ul> |                                                  |
| <b>Seoul Donghaeng Project Mentor</b><br><i>Manna Regional Children’s Center</i>                                                                         | <i>Sep 2024 – Dec 2024</i><br>Seoul, South Korea |
| <ul style="list-style-type: none"><li>Conducted mathematics mentoring and high school entrance guidance for multicultural 8th-grade students.</li></ul>  |                                                  |

SKILLS, CERTIFICATIONS & LANGUAGES

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|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <ul style="list-style-type: none"><li><b>Technical Skills:</b> Python, PyTorch, MATLAB, L<sup>A</sup>T<sub>E</sub>X, Git, Linux / Ubuntu</li></ul> |                           |
| <ul style="list-style-type: none"><li><b>SQL Developer (SQLD)</b>, Korea Data Agency (K-DATA)</li></ul>                                            | <i>Dec 2024</i>           |
| <ul style="list-style-type: none"><li><b>Advanced Data Analytics Semi-Professional (ADsP)</b>, Korea Data Agency (K-DATA)</li></ul>                | <i>Nov 2024</i>           |
| <ul style="list-style-type: none"><li><b>English:</b> TOEIC (835)</li></ul>                                                                        | <i>Aug 2022 (Expired)</i> |

ADDITIONAL INFORMATION

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|----------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <ul style="list-style-type: none"><li><b>Military Service:</b> Republic of Korea Army (Sergeant, Honorably Discharged)</li></ul> | <i>Feb 2023 – Aug 2024</i> |
| <ul style="list-style-type: none"><li><b>Interests:</b> Formula 1 (Red Bull Racing / Max Verstappen)</li></ul>                   |                            |